

Preprocessing Steps & Business Questions with Visualization

Plugging Into the Future: An Exploration of Electricity Consumption Patterns

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1. Dataset Overview

The project uses a state-wise electricity consumption dataset (Consumption.csv) containing 16,600 records with the following columns: States, Regions, latitude, longitude, Dates, and Usage. The data spans multiple Indian states across 2019-2020 and is grouped into five national grid regions (NR, SR, ER, WR, NER).

2. Data Preprocessing Steps

1. Removed duplicate rows and records with null/missing Usage values.
2. Parsed the Dates column into a standard datetime format and derived Month and Year fields for time-series grouping.
3. Standardized State and Region name spellings/casing to ensure consistent grouping and joins.
4. Validated latitude/longitude pairs for each state to support accurate geographic (map) visualizations.
5. Aggregated Usage by State, Region, Month, and Year to build the summary tables used in the dashboards.
6. Checked Usage values for outliers/anomalies before charting to avoid skewed axis scaling.
7. Loaded the cleaned, aggregated data into Tableau as the primary data source for all dashboards and the story.

3. Business Questions

- How does electricity consumption trend on a month-to-month basis across the year?
- Which states or regions consume the most electricity, and how do they compare?
- Are there visible seasonal peaks (e.g. summer cooling, winter heating) in demand?
- How does consumption compare year-over-year, and what growth trends are visible?
- Can stakeholders filter and drill down into consumption by state, month, and year?

4. Initial Visualizations Answering These Questions

The visualizations below (from the project's Tableau dashboards) were used to explore and answer the business questions above: a monthly trend line for question 1 and 3, a state-wise

ranked bar chart and heatmap for question 2, and a region-wise breakdown supporting filtering (question 5).



S.No	Parameter	Details
4	Calculation Fields Used	Monthly Consumption Total, Average Consumption by Region, Year-over-Year Growth %, Peak Month Indicator
5	Dashboard Design	No. of Visualizations / Graphs — 8: Monthly Consumption Trend, State-wise Consumption Map, Top Consuming States, Consumption by Region, Seasonal Peak Analysis, Year-over-Year Growth, Peak vs. Off-Peak Comparison, Consumption Distribution by Month
6	Story Design	No. of Visualizations / Graphs — 5: National Trend Overview, Regional Comparison, Seasonal Peaks, State Rankings, Year-over-Year Growth

7. Results

7.1 Output Screenshots

Dashboard 1 — "Monthly Electricity Consumption Overview" presents the total monthly consumption trend for the selected year alongside a breakdown by region, allowing users to immediately spot seasonal peaks (typically summer and winter months) and troughs.

Dashboard 2 — "State-wise Consumption Comparison" uses a geographic heatmap and ranked bar chart to show which states/regions consume the most electricity, supporting regional planning and equitable resource allocation.

Story — "Understanding Electricity Consumption Patterns" walks the viewer through national trends, regional comparisons, seasonal peaks, and year-over-year growth in a guided narrative, making the insights accessible to non-technical stakeholders.

Electricity Consumption Dashboard

